

FORECASTING LAHORE'S AIR QUALITY THROUGH WEATHER INDICATORS

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Turning open data into early warning systems for cleaner cities.



INTRODUCTION

Lahore ranks among the world's most polluted cities, with $PM_{2.5}$ levels often exceeding safe limits. This study analyzes **28,000+ hourly observations (2021–2024)** to explore how weather affects air pollution and to forecast smog levels using machine learning. The goal is to support **data-driven alerts and urban resilience policies**.

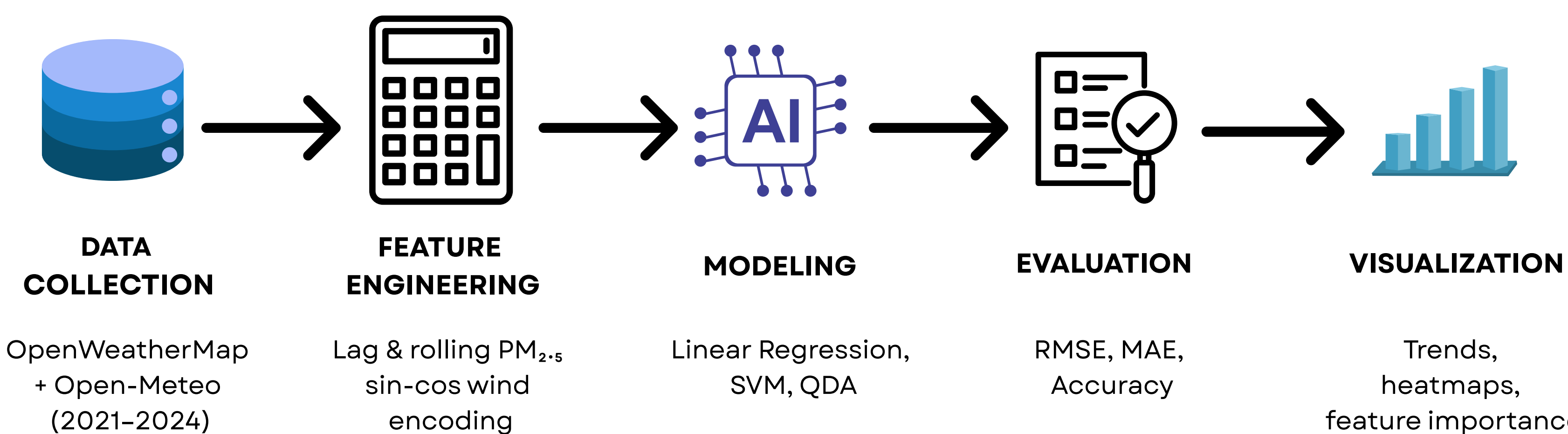
AIMS & OBJECTIVES

- Examine Lahore's air quality and smog seasonality.
- Identify meteorological factors affecting $PM_{2.5}$.
- Develop machine learning models for short-term forecasting.
- Inform early warning systems and health policy.

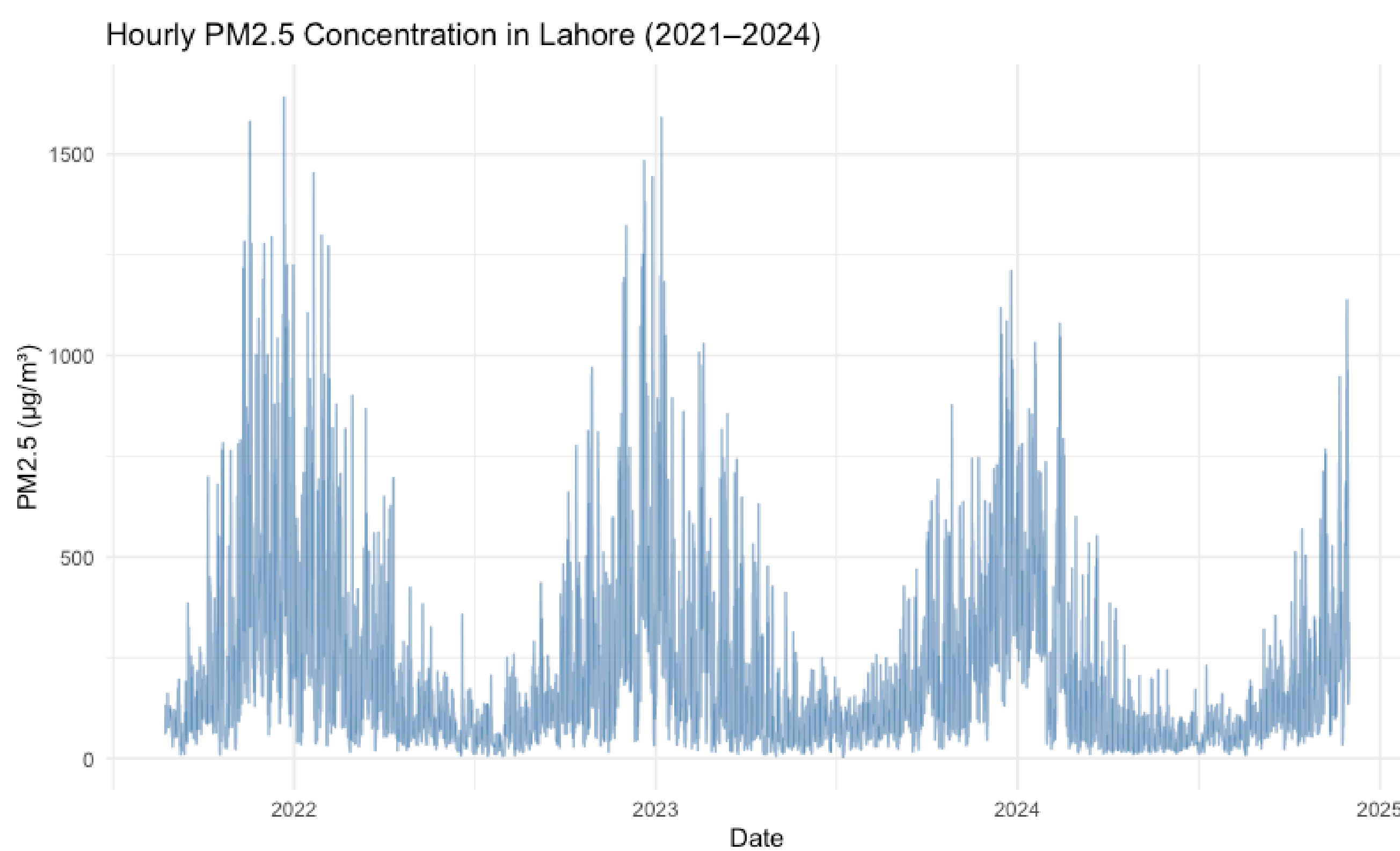
DATA SOURCES

- APIs:** OpenWeatherMap (air quality), Open-Meteo (weather data).
- Period:** Aug 2021–Dec 2024.
- Observations:** 28,269 hourly (training) + 4,368 (testing).
- Variables:** $PM_{2.5}$, gases, humidity, temperature, pressure, wind, radiation. - **City:** Lahore.

METHODOLOGY

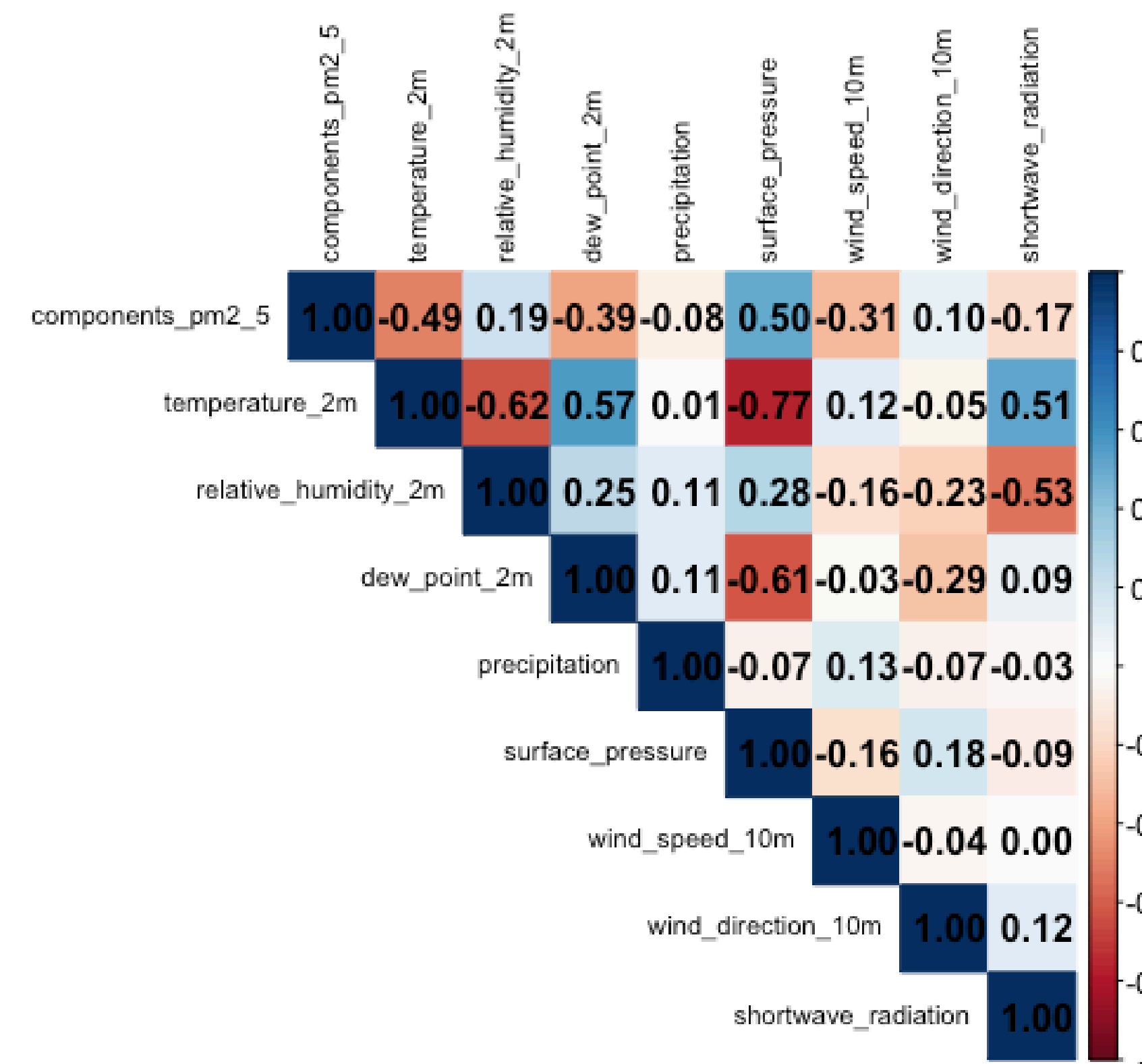


EXPLORATORY DATA ANALYSIS



Hourly $PM_{2.5}$ shows repeating winter peaks and cleaner monsoon months

Correlation Between Weather Variables and $PM_{2.5}$



Temperature, pressure, humidity, wind, radiation showed meaningful relationships with $PM_{2.5}$

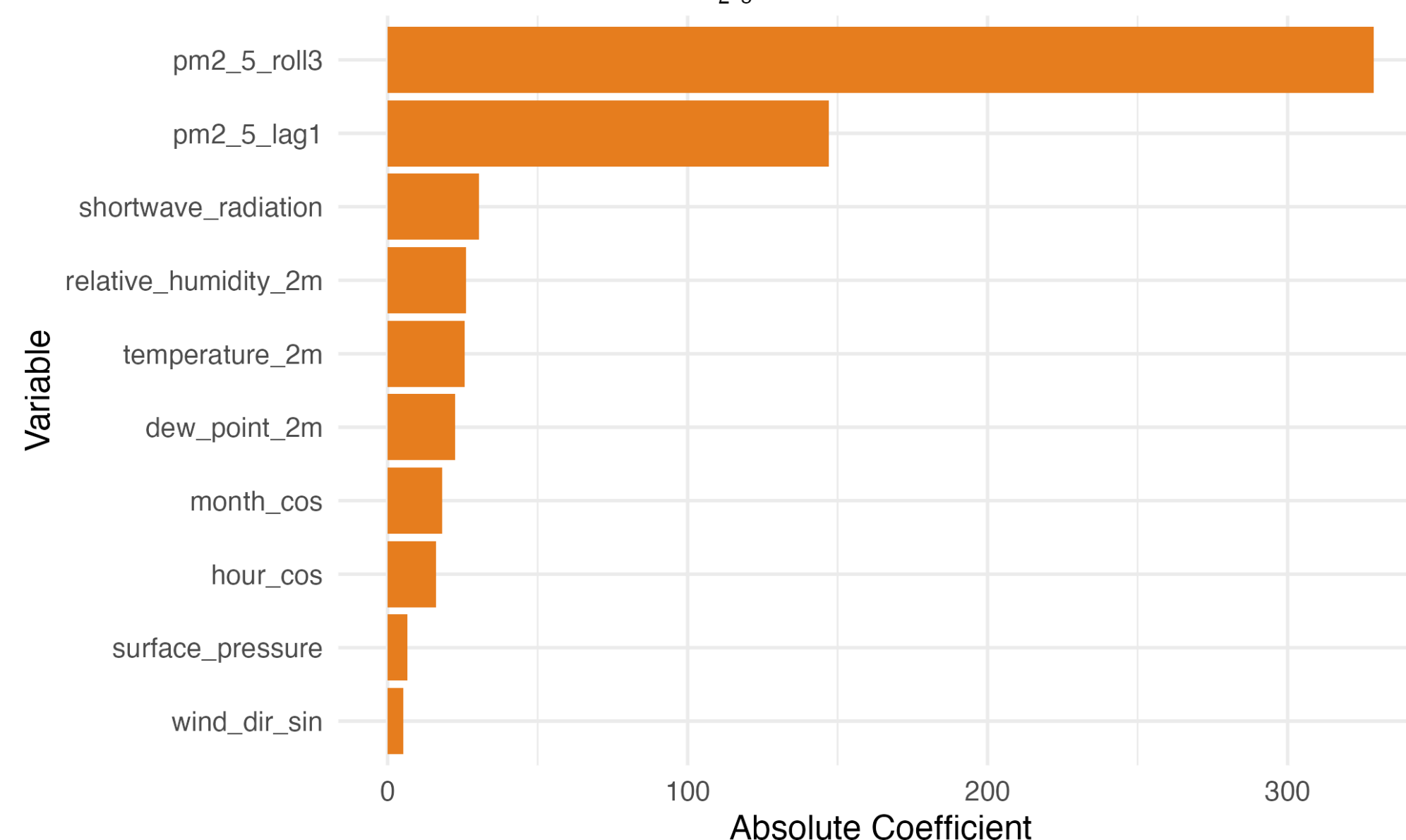
MODELING & RESULTS

We trained three models to forecast $PM_{2.5}$ levels. **Linear Regression** captured general trends, while **SVM (RBF)** best modeled nonlinear smog patterns (RMSE = 173.7, MAE = 115.8). **QDA** classified AQI categories but had lower accuracy due to overlap.

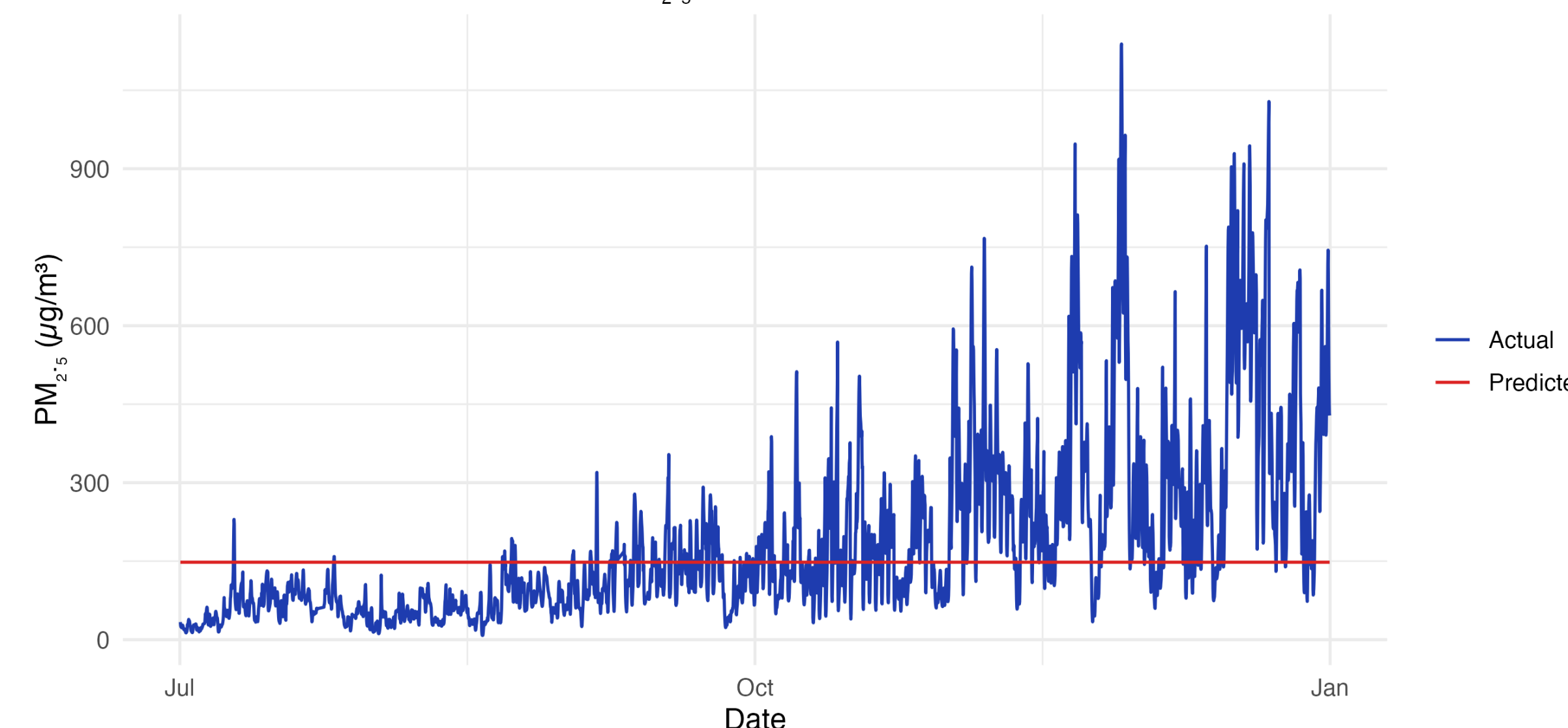
Model Performance Summary

Model	Type	Output	Metric	Key Insight
Linear Regression	Regression	Next-hour $PM_{2.5}$	NRMSE = 22.8%, MAE = 150	Captures trend but misses peaks
SVM (RBF)	Regression	Next-hour $PM_{2.5}$	NRMSE = 18.9%, MAE = 115.8	Best performance; models nonlinear smog dynamics
QDA	Classification	AQI category (Good–Hazardous)	Classification Error = 79.4%	High error due to overlapping pollution categories

Top Predictors of $PM_{2.5}$ (Linear Model)



Actual vs Predicted Next-Hour $PM_{2.5}$ (Lahore, Jul–Dec 2024)



SVM achieves lowest error; QDA = 79.4% Error

Past $PM_{2.5}$, humidity, sunlight, and temperature drive predictions

SVM captures hourly $PM_{2.5}$ variation accurately (for training data)

KEY FINDINGS

- Smog peaks every December–January, lowest in July–August.
- Cold, dry, high-pressure air traps pollutants near surface level.
- SVM model provides reliable short-term air quality forecasts.
- Weather patterns can predict hazardous air quality hours ahead.

Smog in Lahore is predictable.

Data-driven forecasts can save lives.

POLICY IMPACT

Predictive modeling can:

- Trigger real-time smog alerts and public advisories.
- Guide traffic and industrial restrictions on high-risk days.
- Support evidence-based environmental planning.

Future Work: - Expand to daily & multi-city forecasts. - Create live dashboards for Lahore. - Test LSTM for longer-term predictions.